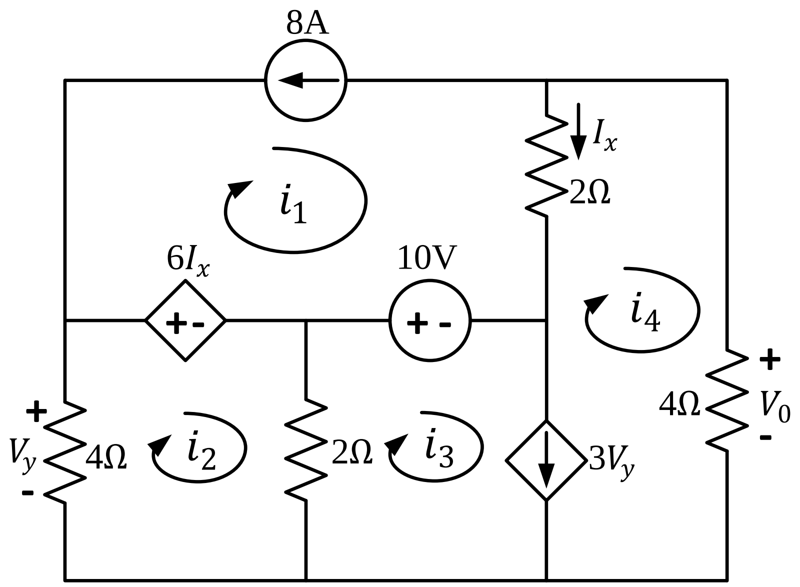
EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Fall 2021 Semester

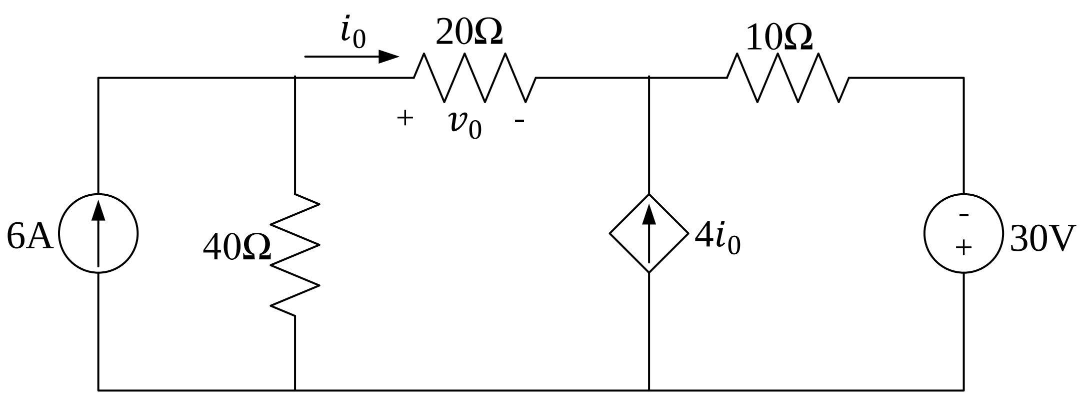
Course: CSE 209 Electrical Circuits, Section-3
Instructor: Musharrat Khan, Senior Lecturer, CSE Department
Full Marks: 40 (20 will be counted for final grading)
Time: 1 Hour and 25 Minutes (Including submission)

Note: There are FIVE questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

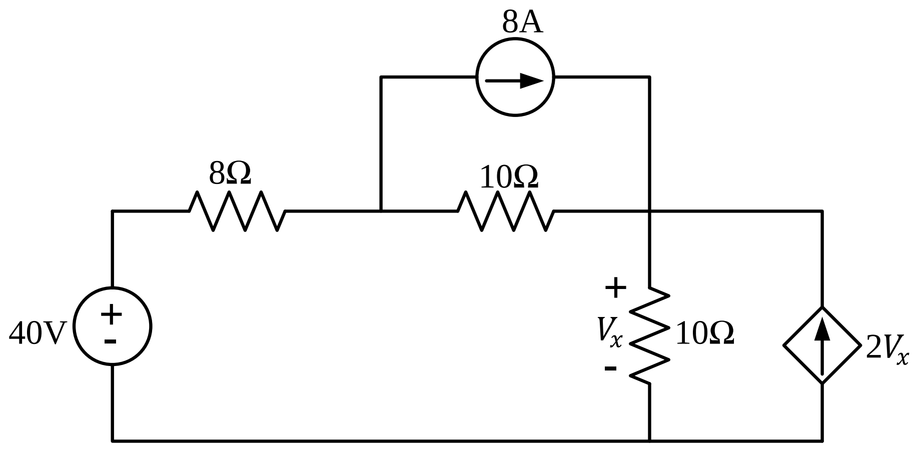
1. Using mesh currents indicated in the circuit, **determine** i_1, i_2, i_3 and i_4 in the following circuit. Also **determine** V_0. [CO2,C4, Mark:12]



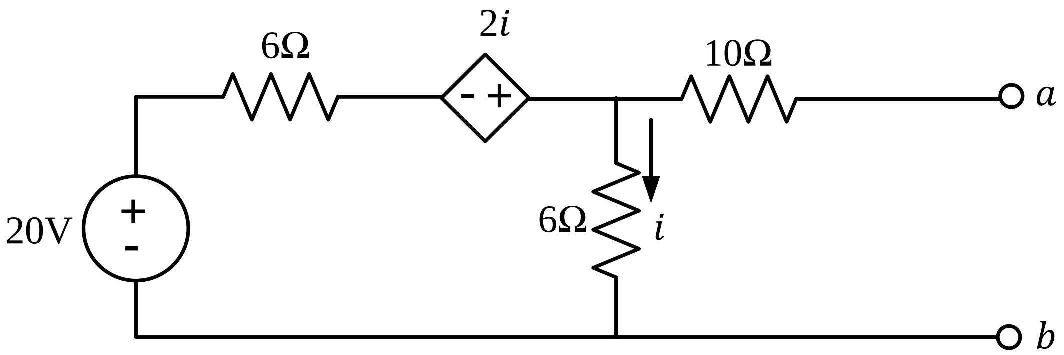
2. Using superposition theorem, **find** v_0 and i_0 in the following circuit. [CO2,C4, Mark: 8]



3. **Use** most effective source transformation to determine the voltage V_x in the following circuit. [CO2,C4, EP1, Mark: 6]



4. **Determine** the Norton's equivalent of the following circuit with respect to terminals a and b. [CO2,C4, EP1,EP2, Mark: 8]



5. **Determine** the value of R_L for maximum power transfer to the load of the following circuit. **Calculate** the maximum power. [CO2,C4, EP1,EP2, Mark: 6]